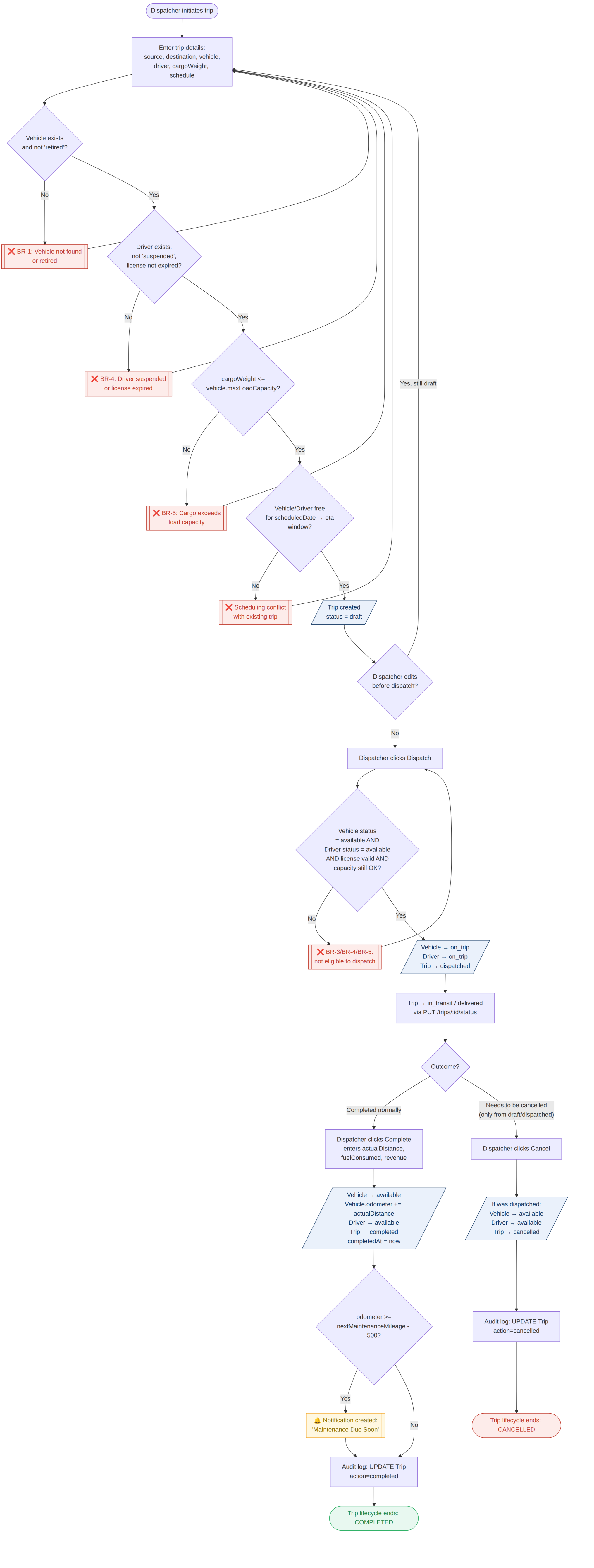
TransitOps — Trip Lifecycle Workflow

End-to-end process flow: creation → dispatch → in-transit → completion/cancellation, with enforced business rules (BR-1, BR-3, BR-4, BR-5) and scheduling-conflict checks


Dispatcher initiates trip
Enter trip details: source, destination, vehicle, driver, cargoWeight, schedule
Vehicle exists and not 'retired'?
No
Yes
❌ BR-1: Vehicle not found or retired
Driver exists, not 'suspended', license not expired?
No
Yes
❌ BR-4: Driver suspended or license expired
cargoWeight <= vehicle.maxLoadCapacity?
No
Yes
❌ BR-5: Cargo exceeds load capacity
Vehicle/Driver free for scheduledDate → eta window?
No
Yes
❌ Scheduling conflict with existing trip
Trip created status = draft
Yes, still draft
Dispatcher edits before dispatch?
No
Dispatcher clicks Dispatch
Vehicle status = available AND Driver status = available AND license valid AND capacity still OK?
No
Yes
❌ BR-3/BR-4/BR-5: not eligible to dispatch
Vehicle → on_trip
Driver → on_trip
Trip → dispatched
Trip → in_transit / delivered via PUT /trips/:id/status
Outcome?
Completed normally
Needs to be cancelled (only from draft/dispatched)
Dispatcher clicks Complete enters actualDistance, fuelConsumed, revenue
Dispatcher clicks Cancel
Vehicle → available
Vehicle.odometer += actualDistance
Driver → available
Trip → completed
completedAt = now
If was dispatched:
Vehicle → available
Driver → available
Trip → cancelled
odometer >= nextMaintenanceMileage - 500?
Audit log: UPDATE Trip action=cancelled
Yes
No
🔔 Notification created: 'Maintenance Due Soon'
Trip lifecycle ends: CANCELLED
Audit log: UPDATE Trip action=completed
Trip lifecycle ends: COMPLETED